LEVERAGE v3

Elite Procurement AI Operating System

Master Schema · Agent Architecture · Development Roadmap

Revision 2 — Updated API Registry (Search, News, Commodity)

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Commercial Operations · Procurement Practice

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Revision Summary (v1 → v2)

This revision replaces three external paid/freemium APIs with superior free alternatives discovered through systematic research. All changes are backward-compatible — no schema table additions required. The API Registry (Section 3) is fully updated. Task list adds two new tasks (P1-15b, P1-23). Tech stack updated throughout.

Replaced	Replacement Stack	Used By
Perplexity Sonar (pay-per-call)	SearXNG self-hosted (unlimited, $0) + DDGS fallback + Serper.dev reserve + Exa.ai semantic	Search / web grounding for all agents, NL co-pilot, category intel
NewsAPI.ai (2,000 lifetime, then paid)	NewsData.io (2,000/day free, resets) + DDGS news + RSS feedparser + Trafilatura (full text)	Supplier risk news monitoring for Agent 5B
CommodityPriceAPI (2,000/mo freemium)	yfinance (20+ futures, no key) + FRED PPI series + EIA energy + World Bank Pink Sheet	Commodity benchmarks for Agent 3A and live benchmark injection

SECTION 1: DATABASE SCHEMA

28 Tables Total — 20 Existing (v2) + 8 New (v3) — No Changes from v1

> Schema unchanged from Revision 1. The API replacements are service-layer changes only — no new database tables required. All 28 tables remain as specified.

1.1 Existing Tables (v2, Preserved)

#	Table Name	Purpose
1	`engagements`	Core engagement record. Every other table FK to engagement_id. Includes branding fields for white-label client deliverables.
2	`categories`	UNSPSC-aligned taxonomy nodes (L1/L2/L3). Global + engagement-specific categories.
3	`spend_records`	Core AP transaction table. Every line item. Drives all engine computations.
4	`savings_initiatives`	Initiative pipeline with scoring, status, probability, risk-adjusted targets.
5	`scenarios`	Alternative savings scenarios per initiative (assumption variants).
6	`data_imports`	File upload tracking with column mapping and staging data.
7	`realization_entries`	Actual savings bookings by date. Feeds realized tracker.
8	`cash_flow_phasing`	Monthly cash flow schedule per initiative.
9	`cleansing_audit_log`	Audit trail of all data changes during cleansing.
10	`supplier_mappings`	Canonical name mappings for normalized suppliers.
11	`category_rules`	User-defined GL/supplier/description to category mapping rules.
12	`assumption_benchmarks`	Engagement-level benchmark overrides per lever/category.
13	`tariff_impacts`	Computed tariff exposure per supplier/category.
14	`contracts`	Contract metadata: value, dates, payment terms, escalation clauses.
15	`procurement_maturity_assessments`	8-dimension Hackett maturity scores with evidence.
16	`tariff_sourcing_scenarios`	Alternative sourcing country analysis with net benefit calc.
17	`monte_carlo_runs`	Stored Monte Carlo outputs: P10/P50/P90 by initiative and phase.
18	`category_strategy`	Kraljic-derived sourcing strategy per L2 category.
19	`spend_summaries`	Pre-aggregated spend rollups for performance (cache table).
20	`fx_rates`	FX rates per currency per engagement for multi-currency normalization.

1.2 New Tables for v3 (8 Net-New)

Table 21: agent_jobs [NEW]

Field	Type	Key	Description
`id`	`INTEGER`	PK	Auto-increment

Field	Type	Key	Description
engagement_id	INTEGER	FK	FK → engagements.id (nullable for cross-engagement jobs)
agent_id	TEXT NOT NULL		'agent_0' | 'agent_1a' | 'agent_3a' | 'agent_6' | etc.
job_type	TEXT NOT NULL		'data_ingest' | 'market_refresh' | 'deliverable_gen' | 'supplier_risk_scan' | 'contract_extract' | 'news_scan' | 'commodity_refresh'
status	TEXT NOT NULL		'queued' | 'running' | 'complete' | 'failed' | 'cancelled'
progress_pct	INTEGER		0–100 progress integer for SSE streaming to frontend
progress_message	TEXT		Current step description shown in UI progress bar
input_json	TEXT		JSON: job input parameters
output_json	TEXT		JSON: job result payload (file path, summary, metrics)
error_message	TEXT		Error details if status=failed
retry_count	INTEGER		Number of retry attempts. Max 3 with exponential backoff
started_at	TEXT		ISO-8601 timestamp
completed_at	TEXT		ISO-8601 timestamp
created_at	TEXT		ISO-8601 timestamp

Table 22: supplier_risk_profiles [NEW]

Field	Type	Key	Description
id	INTEGER	PK	Auto-increment
engagement_id	INTEGER NOT NULL	FK	FK → engagements.id
supplier_name	TEXT NOT NULL		Canonical normalized supplier name
sec_cik	TEXT		SEC Central Index Key (public companies only)
altman_z_score	REAL		Altman Z-score: <1.81=distress, 1.81–2.99=grey, >2.99=safe
revenue_trend	TEXT		'growing' | 'declining' | 'stable' (derived from 10-K YoY)
leverage_ratio	REAL		Total debt / EBITDA from latest annual filing
financial_risk_level	TEXT		'low' | 'medium' | 'high' | 'critical'
news_sentiment_score	REAL		-1.0 (very negative) to +1.0 (very positive). Derived from full article text via Trafilatura + Claude
news_risk_flags	TEXT		JSON array: ['labor_dispute','regulatory_action','bankruptcy_risk','exec_departure','ma_activity','cyber_incident','recall','sanctions']
latest_news_headline	TEXT		Most recent risk-relevant headline
latest_news_url	TEXT		Source URL. Trafilatura fetches full text from this URL for Claude analysis

Field	Type	Key	Description
article_confidence	TEXT		'high' (full text extracted) | 'medium' (archive fallback) | 'low' (snippet only). Set by ArticleExtractorService
ofac_match	INTEGER		0/1: found on OFAC SDN sanctions list
sam_exclusion	INTEGER		0/1: found on SAM.gov debarment/exclusions list
overall_risk_score	REAL		0–100 composite risk score (feeds Kraljic supply risk dimension)
risk_narrative	TEXT		Claude-generated 2–3 sentence risk summary based on full article text
last_refreshed_at	TEXT		ISO-8601 timestamp of last full data pull

Table 23: market_data_cache [NEW]

Field	Type	Key	Description
id	INTEGER	PK	Auto-increment
data_source	TEXT NOT NULL		'fred' | 'bls' | 'eia' | 'yfinance' | 'world_bank_pink' | 'drewry' | 'ecb_fx' — NOTE: 'commodity_api' removed
series_id	TEXT NOT NULL		Source-specific series ID. yfinance: 'GC=F','HG=F','CL=F' etc. FRED: 'CPIAUCSL','WPU10' etc. EIA: 'PET.RWTC.W'
series_name	TEXT		Human-readable name (e.g. 'Gold Futures (COMEX)', 'CPI All Urban Consumers')
category_tag	TEXT		'labor' | 'commodity_metal' | 'commodity_energy' | 'commodity_ag' | 'freight' | 'macro' | 'fx' | 'ppi'
value	REAL		Latest data point value
unit	TEXT		'$/oz' \| '$/barrel' | '$/MMBtu' \| '$/TEU' | 'index' | '% YoY'
period	TEXT		Period of latest value (e.g. '2026-03-21' for yfinance, '2026-02' for FRED monthly)
yoy_change_pct	REAL		Year-over-year percentage change
mom_change_pct	REAL		Month-over-month percentage change
raw_json	TEXT		Full API response or yfinance DataFrame row serialized as JSON
fetched_at	TEXT		ISO-8601 timestamp of last fetch
ttl_hours	INTEGER		Cache TTL: yfinance=1h, FRED=24h, BLS=24h, EIA=12h, World Bank=168h (weekly), FX=4h, Drewry=168h

Table 24: contract_extractions [NEW]

Field	Type	Key	Description
id	INTEGER	PK	Auto-increment
engagement_id	INTEGER NOT NULL	FK	FK → engagements.id
contract_id	INTEGER	FK	FK → contracts.id (populated after matching, confidence

Field	Type	Key	Description
			> 0.7)
file_name	TEXT NOT NULL		Original uploaded filename
file_path	TEXT		Server path to stored file
extraction_status	TEXT		'pending' | 'processing' | 'complete' | 'failed'
supplier_name_extracted	TEXT		Claude-extracted counterparty name
contract_value_extracted	REAL		Claude-extracted annual contract value
start_date_extracted	TEXT		Claude-extracted effective start date
end_date_extracted	TEXT		Claude-extracted expiry/end date
payment_terms_extracted	TEXT		Claude-extracted payment terms
auto_renewal_extracted	INTEGER		0/1: Claude-detected auto-renewal clause present
escalation_clause_extracted	TEXT		Verbatim escalation clause text if found
key_clauses_json	TEXT		JSON: {termination_rights, SLA, liability_cap, exclusivity, notice_period}
risk_flags_json	TEXT		JSON array of identified risk clauses
confidence_score	REAL		0.0–1.0 extraction confidence (>0.7 auto-creates contracts row)
raw_text	TEXT		Full extracted plain text of the contract document
claude_summary	TEXT		Claude-generated 3-sentence contract brief
extracted_at	TEXT		ISO-8601 timestamp

Tables 25–28: Additional New Tables

#	Table Name	Purpose
25	deliverable_outputs	Generated file tracking: type (pptx/docx/xlsx/pdf), engagement, file path, timestamp, Claude model version, token count
26	watchlist_alerts	Active alerts: contract_expiry, supplier_distress, commodity_spike, savings_at_risk, ofac_match. Severity + acknowledged/resolved timestamps
27	copilot_sessions	NL co-pilot conversation history per engagement. Message history JSON, tool calls made, data sources referenced, session name
28	portfolio_snapshots	Cross-engagement aggregate KPIs computed nightly. Total pipeline $, realized $, avg savings rate, peer benchmarks (anonymized)

SECTION 2: AGENT ARCHITECTURE

14 Agents · Unchanged from Revision 1 — See API Registry for updated data sources

Agent architecture is unchanged from Revision 1. The web search backbone for Agents 2, 5, 7, and 10 now uses SearXNG (primary) → DDGS (fallback) → Serper.dev (Google-quality reserve) → Exa.ai (semantic). Agent 5B supplier news now uses NewsData.io → DDGS news → RSS, with Trafilatura extracting full article text from URLs. Agent 3A commodity feeds now use yfinance → FRED PPI → EIA → World Bank.

#	Agent	Type	Build	Phase	Key Data Sources / Notes
0	Master Orchestrator	**AI-NATIVE**	New hire	Phase 1	All task routing, quality gates, conflict resolution. Web search via SearXNG for grounding.
1A	Data Ingestion & Normalization	**EXTEND**	Extend	Phase 1	Auto column mapping, dedup, FX, GL standardization.
1B	Data Quality & Audit	**EXTEND**	Extend	Phase 1	Statistical outlier detection, completeness scoring, Claude quality narrative.
1C	Spend Classification	**EXTEND**	Extend	Phase 1	6-strategy UNSPSC + Claude LLM fallback. SearXNG used for ambiguous supplier lookups.
1D	Contract Intelligence	**NEW**	New hire	Phase 1	PDF/docx ingestion, Claude clause extraction, expiry alerts. Trafilatura not used here.
2	Spend & Category Intelligence	**EXTEND**	Extend	Phase 1	Spend flags + Claude brief. SEC EDGAR financial health. SearXNG for supplier web lookups.
3	Live Market Intelligence	**NEW**	Critical	Phase 1	3A: yfinance+FRED+EIA+World Bank commodities. 3B: BLS OES labor. 3C: Drewry WCI freight. 3D: FRED+OECD macro.
4	Savings & Financial Modeling	**EXTEND**	Extend	Phase 1	Scoring + live signals. EBITDA bridge. USITC tariff lookup. FRED for risk-free rate.
5	Supplier Risk Monitor	**NEW**	New hire	Phase 2	5A: SEC EDGAR Altman Z. 5B: NewsData.io + DDGS + RSS → Trafilatura full text → Claude risk flags. 5C: HHI. 5D: OFAC+SAM.gov.
6	Deliverable Generation Engine	**NEW**	Critical	Phase 1	1-click steerco (pptx), ODD memo (docx), Excel model, PDF. Claude authors all narrative.
7	NL Co-pilot	**AI-NATIVE**	Critical	Phase 1	All engine tools exposed to Claude. SearXNG for live web grounding. Exa.ai for semantic supplier research.
8	Async Job Queue	**INFRA**	New hire	Phase 1	BullMQ + Redis. SSE streaming. Retry logic. All agent tasks run here.
9	MD Portfolio Command Center	**NEW**	New hire	Phase 3	Cross-engagement KPIs, team utilization, peer benchmarks, PE sponsor rollup.
10-13	Category Advisor, 100-Day, FX, Alerts	**EXTEND/ NEW**	Phase 2-3	Phase 2-3	Exa.ai semantic search for competitor supplier intel. ECB FX. nodemailer alerts.

SECTION 3: API REGISTRY — UPDATED

Complete source map with all replacements incorporated · March 2026

All three replacements reflected below. Deleted: Perplexity Sonar, NewsAPI.ai, CommodityPriceAPI. Added: SearXNG, DDGS, Serper.dev, Exa.ai, NewsData.io, Trafilatura, feedparser, yfinance, World Bank Pink Sheet. Cost column reflects true cost at Leverage v3 operating volumes.

3.1 Web Search Stack (replaces Perplexity Sonar)

API	Provider	Cost	Endpoint	Auth	Quota	Usage / Notes
SearXNG	Self-hosted metasearch	$0 forever	`http://localhost:8888/search?format=json`	None (self-hosted)	Unlimited	PRIMARY for all agents. Docker container on same box. Aggregates 70+ engines (Google, Bing, DDG, Brave). No rate limit. Private.
DDGS (Python)	DuckDuckGo unofficial	$0 forever	`ddgs.text() / ddgs.news()`	None (pip install ddgs)	Soft limit	FALLBACK when SearXNG rate-limited. No key, no signup. Handles news search too via ddgs.news(). Unofficial — can break.
Serper.dev	Google SERP API	$0 (2,500 one-time)	`https://google.serper.dev/search`	Register: serper.dev	2,500 free	RESERVE for high-value queries needing true Google results. Competitor intel, specific company lookups. After 2,500: $1/1k.
Exa.ai	Neural semantic search	$10 credit no expiry	`https://api.exa.ai/search`	Register: exa.ai	~2,000 searches	SEMANTIC queries only. 'Find suppliers of industrial gases similar to Air Products.' Agent 10 category intel. After credit: $5/1k.

3.2 News & Article Intelligence Stack (replaces NewsAPI.ai)

Source	Type	Cost	Endpoint / Call	Auth	Quota	Role in Pipeline
NewsData.io	News aggregator API	$0 (200 cred/day)	`https://newsdata.io/api/1/latest?apikey=KEY&q=QUERY`	Register: newsdata.io	2,000 articles/day, resets	PRIMARY news source. Returns headline + snippet + URL. 77 languages, 151 countries. 12h delay on free tier. Commercial use OK. No CC.
DDGS news	DuckDuckGo news	$0 forever	`ddgs.news(query, max_results=20)`	None (pip install ddgs)	Unlimited (soft)	FALLBACK when NewsData.io daily credit exhausted. Real-time. Also used for on-demand co-pilot news queries.
RSS feedparser	Trade journal feeds	$0 forever	`feedparser.parse(RSS_URL)`	None (pip install feedparser)	Unlimited	Supply Chain Dive, Spend Matters, FreightWaves, Reuters Commodities, CSCMP. Parsed hourly. Headline + summary. No delay.
Trafilatura	Full article extractor	$0 forever	`trafilatura.fetch_url(url) + .extract()`	None (pip install trafilatura)	Unlimited	EXTRACTS FULL TEXT from NewsData.io article URLs. Apache 2.0. #1 ranked open-source extractor (HuggingFace, IBM, Stanford use it). Fallback chain: direct → archive.ph → snippet.
archive.ph	Article archive	$0 forever	`https://archive.ph/{url}`	None	Unlimited	Trafilatura fallback for paywalled articles (Bloomberg, FT, WSJ). Trafilatura fetches archive.ph URL instead of original.

Article extraction flow: NewsData.io returns URL → Trafilatura.fetch_url(url) + .extract() → full article body passed to Claude for risk classification. Confidence field: 'high' (full text), 'medium' (archive.ph fallback), 'low' (snippet only). Stored in supplier_risk_profiles.article_confidence. For procurement risk monitoring, ~85–90% of articles from trade press + regional business news are fully extractable without paywall issues.

3.3 Commodity Price Stack (replaces CommodityPriceAPI)

Source	Provider	Cost	Endpoint / Call	Auth	Quota	Coverage / Use Case
yfinance	Yahoo Finance futures	$0 forever	`yf.download('GC=F CL=F HG=F ...')`	None (pip install yfinance)	Unlimited	PRIMARY for real-time spot prices. 20+ CME/COMEX/NYMEX futures: metals (GC=F, HG=F, ALI=F, SI=F, PA=F, PL=F, NI=F), energy (CL=F, BZ=F, NG=F), ag (ZW=F, ZC=F, ZS=F, CT=F, KC=F, SB=F), lumber (LBS=F), steel (HR=F). One call, all tickers.
FRED API	US Federal Reserve	$0 (free key)	`https://api.stlouisfed.org/fred/series/observations`	Register: fred.stlouisfed.org	840k+ series, unlimited	PPI commodity series: WPU10 (metals), WPU06 (chemicals & plastics), WPU05 (lumber), WPU091 (energy). Monthly. Also: CPIAUCSL, FEDFUNDS, GS10, ECI. Used for inflating historical benchmarks.
EIA API	US Energy Info Admin	$0 (free key)	`https://api.eia.gov/v2/`	Register: eia.gov/opendata	Unlimited	AUTHORITATIVE for energy: WTI weekly (PET.RWTC.W), Henry Hub nat gas (NG.RNGWHHD.W), regional electricity rates. More reliable than yfinance for energy; use as primary energy source.
World Bank Pink Sheet	Global commodity data	$0 forever	`thedocs.worldbank.org/... CMO-Historical-Data-Monthly.xlsx`	None	Monthly Excel download	70+ commodities, 60yr history. Agricultural inputs, fertilizers, tropical commodities not on US exchanges. Auto-download monthly xlsx. Parse with openpyxl. Best for: rubber, cocoa, palm oil, phosphate.
ECB FX API	Euro FX rates	$0 forever	`https://data-api.ecb.europa.eu/service/data/EXR/`	None	Unlimited	40+ currency pairs to EUR. Used by Agent 12 FX service for spend normalization. TTL: 4 hours.

3.4 All Other APIs (unchanged from Revision 1)

API	Provider	Cost	Endpoint	Auth	TTL	Use Case
FRED API	Federal Reserve	$0 (free key)	`https://api.stlouisfed.org/fred/`	Register	24h	840k+ economic series. CPI, PPI, Fed Funds, ECI, 10Y Treasury, GDP.
BLS API v2	Bureau of Labor Statistics	$0 (free key)	`https://api.bls.gov/publicAPI/v2/`	Register: bls.gov/developers	24h	OES wages by occupation + geography, CPI, QCEW employment cost. Agent 3B.
SEC EDGAR	US SEC filings	$0 (no key)	`https://data.sec.gov/`	None	24h	10-K/10-Q financials for public suppliers. 8-K material events. Agent 5A Altman Z-score.
World Bank Open Data	Global development	$0 (no key)	`https://api.worldbank.org/v2/`	None	24h	GDP, inflation, trade for 196 countries. Macro context for engagement geography.
OECD API	38-country economics	$0 (no key)	`https://stats.oecd.org/SDMX-JSON/`	None	24h	GDP, unemployment, CPI, trade. Macro for non-US engagements.
EIA API	US energy data	$0 (free key)	`https://api.eia.gov/v2/`	Register: eia.gov	12h	WTI, nat gas, electricity. Primary energy price source for Agent 3A.
OpenCorporates	200M+ companies	$0 (free tier)	`https://api.opencorporates.com/`	Free: 50/day	24h	Company ownership, officers, jurisdictions. Supplier due diligence.

API	Provider	Cost	Endpoint	Auth	TTL	Use Case
OFAC SDN	US sanctions list	$0 (no key)	`https://www.treasury.gov/ofac/downloads/`	None	Daily dl	Bulk XML download. Supplier sanctions screening. Agent 5D.
SAM.gov	US federal entity registry	$0 (free key)	`https://api.sam.gov/entity-information/`	Register: sam.gov	24h	Entity registration, debarment/exclusions. Agent 5D.
USITC DataWeb	US tariff + trade data	$0 (account)	`https://dataweb.usitc.gov/`	Register	Daily	HTS tariff rates by product code and country. Agent 4C tariff modeler.
Claude API	Anthropic AI	Your key	`https://api.anthropic.com/v1/messages`	Your key	N/A	ALL narrative, synthesis, extraction, co-pilot, risk classification, deliverable authoring.
Drewry WCI	Freight index	$0 (scrape)	`thedocs.worldbank.org (published weekly)`	None	Weekly	World Container Index — 8 trade lanes. Playwright scraper. Agent 3C freight benchmarks.

SECTION 4: DEVELOPMENT TASK LIST — UPDATED

39 Tasks (added P1-15b and P1-23) · All API references updated

PHASE 1 — Foundation (Weeks 1–4)

Goal: Async queue + Python sidecar + NL co-pilot + Steerco deck + Live market feeds + Contract ingestion + SearXNG + Full-text news

ID	Task	Description / Acceptance Criteria	Agent	Effort
INFRA — Schema Migration & Job Queue				
P1-01	**Schema v3 migration**	Add 8 new tables (agent_jobs, supplier_risk_profiles, market_data_cache, contract_extractions, deliverable_outputs, watchlist_alerts, copilot_sessions, portfolio_snapshots). Add DDL to storage.ts. Bump SQLite user_version to 3. Add Drizzle types to schema.ts. Note: market_data_cache.data_source enum includes 'yfinance' and 'world_bank_pink' (replaces 'commodity_api').	Infra	2d
P1-02	**BullMQ job queue setup**	Install bullmq + ioredis. Create server/services/JobQueueService.ts with enqueue(), getStatus(), cancel(). Wire to agent_jobs table. 3-retry exponential backoff. Register queue processor workers. Job types include: news_scan, commodity_refresh, article_extract.	Agent 8	2d
P1-03	**SSE progress streaming**	GET /api/jobs/:id/stream returns Server-Sent Events. Emit {progress_pct, message} on each checkpoint. Frontend JobStatusPanel with progress bar + live message. Polling fallback every 3s if SSE fails.	Agent 8	1d
P1-04	**Python sidecar service**	Create leverage-app/python-sidecar/main.py as FastAPI on port 5001. Endpoints: POST /generate/pptx, /generate/docx, /generate/xlsx, /generate/pdf. Install python-pptx, python-docx, openpyxl, reportlab. Also install trafilatura and feedparser in the same Python environment.	Agent 6	2d
INFRA — SearXNG Search Infrastructure [NEW in Revision 2]				
P1-15b	**SearXNG Docker + WebSearchService**	Deploy SearXNG Docker container on same server as Leverage (port 8888). Enable JSON output format (required, disabled by default). Create server/services/WebSearchService.ts with 4-tier fallback: (1) SearXNG localhost, (2) DDGS via Python subprocess, (3) Serper.dev for highValue=true queries, (4) Exa.ai for semantic=true queries. All agents call WebSearchService.search(query, opts) never APIs directly. Test with supplier risk queries.	Infra	2d
AGENT-07 — NL Co-pilot				
P1-05	**Co-pilot API route**	POST /api/engagements/:id/copilot. Accepts {message, history[]}. Injects full engagement context into Claude system prompt. Web search via WebSearchService (SearXNG primary). Stream response via SSE.	Agent 7	2d
P1-06	**Claude tool definitions**	12 Claude tools: get_spend_summary, get_top_vendors, get_initiative_scores, get_kraljic_matrix, get_financial_model, get_maturity_gap, get_tariff_exposure,	Agent 7	3d

ID	Task	Description / Acceptance Criteria	Agent	Effort
		get_contract_status, run_market_lookup, get_supplier_risk, search_engagement_data (via WebSearchService), generate_deliverable.		
P1-07	**Co-pilot UI panel**	Slide-out chat panel (right side, Cmd+K). Message history, streaming text, tool-call indicators ('Searching market data...'), citation chips. Session save + restore.	Agent 7	3d
P1-08	**Session persistence**	Save copilot_sessions rows. Resume on page reload. MD can name sessions. History sidebar with timestamps.	Agent 7	1d
AGENT-06 — Deliverable Generation Engine				
P1-09	**Steerco deck generator**	POST /api/engagements/:id/generate/steerco. Claude drafts 8-slide narrative. Python sidecar renders branded pptx. Slides: situation, spend waterfall, top 10 initiatives, Kraljic summary, EBITDA bridge, 100-day roadmap, top 3 risks, next steps. Store path in deliverable_outputs.	Agent 6	4d
P1-10	**ODD memo generator**	POST /api/engagements/:id/generate/odd-memo. Claude authors full Word doc: exec summary, methodology, spend findings by L1, initiative pipeline, risk matrix, implementation roadmap. python-docx renders. A&M header/footer.	Agent 6	3d
P1-11	**Excel model exporter**	POST /api/engagements/:id/generate/excel. Tabs: Summary, Initiative Pipeline, Financial Model (NPV/IRR), Savings Tracker, Cash Flow, Assumptions. A&M color scheme. Store in deliverable_outputs.	Agent 6	2d
P1-12	**Deliverable UI page**	/engagements/:id/deliverables. Generate buttons per type with loading state. File download list with timestamp. White-label toggle (swap client branding). Version history.	Agent 6	2d
AGENT-03 — Live Market Intelligence (all APIs updated in Revision 2)				
P1-13	**FRED + BLS data service**	Create server/services/MarketDataService.ts. fetchFRED(seriesId) and fetchBLS(seriesIds[]). TTL-based cache check against market_data_cache. Priority FRED series: CPIAUCSL, WPU10 (metals PPI), WPU06 (chemicals PPI), WPU05 (lumber PPI), FEDFUNDS, GS10, ECI. BLS OES wages by SOC code.	Agent 3	2d
P1-14	**yfinance commodity feed (replaces CommodityPriceAPI)**	fetchCommodities() using yfinance Python subprocess (or node-fetch to Python sidecar endpoint). Download all 20 tickers in one call: metals (GC=F, HG=F, ALI=F, SI=F, NI=F), energy (CL=F, BZ=F, NG=F), ag (ZW=F, ZC=F, ZS=F, CT=F, KC=F, SB=F), lumber (LBS=F), steel (HR=F). Store each to market_data_cache with data_source='yfinance'. Add POST /api/market/commodities endpoint. TTL: 1 hour. Also fetch EIA WTI and nat gas weekly as authoritative energy source (data_source='eia').	Agent 3	2d
P1-15	**Freight index scraper**	Playwright scraper for Drewry WCI weekly table (8 trade lanes). Freightos FBX public page. Run weekly via node-cron. Store to market_data_cache (data_source='drewry', ttl_hours=168). Claude writes 1-paragraph freight market interpretation.	Agent 3	2d
P1-16	**Live benchmark injection**	Modify benchmarks.ts to accept optional market_data override from market_data_cache. CPI YoY → inflate historical benchmarks. PPI WPU10 → adjust metals savings rates. PPI WPU05 → adjust lumber. OES	Agent 3	2d

ID	Task	Description / Acceptance Criteria	Agent	Effort
		wages → update labor benchmarks. yfinance commodity delta vs client purchase price → market vs overpay quantification. Auto-refresh nightly via Agent 8 queue.		
P1-17	**Market Intelligence page**	New page /engagements/:id/market-intel. Live commodity price cards with yfinance data + YoY delta. BLS wage trends by engagement categories. Freight indices. FRED macro rates. Claude-written Market Pulse brief generated daily.	Agent 3	2d
AGENT-01D — Contract Intelligence				
P1-18	**Contract file upload**	POST /api/engagements/:id/contracts/upload. Accept PDF/docx/txt. Validate + store. Create contract_extractions row (status=pending). Enqueue Agent 1D job. Return job_id for SSE tracking.	Agent 1D	1d
P1-19	**Claude contract extraction**	Agent 1D job: extract text via pdf-parse (PDF) or mammoth.js (docx). Claude extracts: supplier, value, dates, payment terms, auto-renewal, escalation clause, SLA, termination rights, liability cap. confidence_score > 0.7 auto-creates contracts row.	Agent 1D	2d
P1-20	**Expiry alert engine**	Nightly cron: query contracts with end_date within 90/60/30 days. Create watchlist_alerts (severity low/medium/critical). Surface badge in nav header.	Agent 1D	1d
AGENT-02 — Spend Intelligence				
P1-21	**SEC EDGAR financial health**	For each public supplier: search EDGAR for CIK, fetch 10-K XBRL facts (revenues, net income, debt). Compute Altman Z-score. Claude writes 2-sentence health summary. Store to supplier_risk_profiles.	Agent 2	2d
P1-22	**Claude spend narrative**	POST /api/engagements/:id/spend-narrative. Claude reads top 5 spend flags + top 10 vendors + L1 distribution. 250-word exec brief with 3 prioritized actions. Cache 24h.	Agent 2	1d
AGENT-05B — News & Article Extraction Pipeline [NEW in Revision 2]				
P1-23	**ArticleExtractorService + news pipeline**	Create server/services/ArticleExtractorService.ts. Three-layer article retrieval: (1) Trafilatura via Python sidecar POST /extract/article {url} → full text, (2) archive.ph fallback if Trafilatura returns <200 chars, (3) NewsData.io snippet fallback. Store confidence level ('high'/'medium'/'low') to supplier_risk_profiles.article_confidence. News query pipeline: NewsData.io for 50+ suppliers → Trafilatura full text → Claude risk classification → supplier_risk_profiles update. Add /extract/article endpoint to Python sidecar (trafilatura.fetch_url + .extract()). Also install feedparser in Python env and add /extract/rss endpoint for trade journal monitoring (Supply Chain Dive, Spend Matters, FreightWaves, Reuters Commodities RSS feeds).	Agent 5	3d

PHASE 2 — Intelligence Layer (Weeks 5–10)

Goal: Full supplier risk suite · Alert engine · FX · Financial model enhancements

ID	Task	Description / Acceptance Criteria	Agent	Effort
AGENT-05 — Supplier Risk Monitor (Full Suite)				
P2-01	**News sentiment monitoring (uses Revision 2 stack)**	Nightly Agent 5B job: query NewsData.io for each top-50 supplier. ArticleExtractorService fetches full text via Trafilatura. Claude classifies: sentiment_score, risk_flags[], severity, evidence_quote. Fallback to DDGS news if NewsData.io daily credit exhausted. RSS feeds monitored separately for trade journal signals. Store all to supplier_risk_profiles.	Agent 5	2d
P2-02	**OFAC + SAM.gov screening**	Download OFAC SDN XML (Treasury bulk download, weekly). SAM.gov exclusions API. Fuzzy match suppliers (0.85 similarity threshold). Set ofac_match=1 / sam_exclusion=1. Auto-create Critical watchlist_alert.	Agent 5	2d
P2-03	**Concentration risk (HHI)**	Herfindahl-Hirschman Index per L2 category. Flag HHI > 2500 (concentrated), > 5000 (highly concentrated). Single-source dependencies. Feed Kraljic supply risk dimension.	Agent 5	1d
P2-04	**Supplier Risk dashboard**	/engagements/:id/supplier-risk. Risk matrix (financial health x news sentiment). Top 10 at-risk suppliers. OFAC/SAM flag indicators. Confidence level badge ('full text' / 'archive' / 'snippet') per supplier.	Agent 5	2d
P2-05	**Risk score into scoring engine**	Wire supplier_risk_profiles.overall_risk_score into initiative scoring engine market dimension. Sole-source + high-risk supplier increases supply_risk_factor.	Agent 5	1d
AGENT-12 — FX & Multi-Currency				
P2-06	**ECB FX live feed**	fetchFXRates() from ECB data API. 40+ currencies → USD. Cache to fx_rates table (source='api'). TTL 4h. Fallback to previous cached rate if API down.	Agent 12	1d
P2-07	**FX exposure analysis**	Spend by currency, total FX exposure by country, volatility risk for major pairs. Claude writes FX risk brief.	Agent 12	1d
AGENT-13 — Alert Engine				
P2-08	**Commodity spike alerts**	Nightly job: for each commodity in market_data_cache (data_source='yfinance' or 'eia'), compare to 30-day average. If MoM change > 5%, find exposed engagement categories, create watchlist_alerts.	Agent 13	1d
P2-09	**Savings at-risk alerts**	Nightly: compare realized_amount vs prorated target. Flag if YTD realization < 50% of expected. Set is_at_risk=1 + create watchlist_alert.	Agent 13	1d
P2-10	**Alert center UI + email**	/engagements/:id/alerts. Grouped by type, color-coded severity. Acknowledge/resolve buttons. nodemailer emails on Critical severity. Nav badge with unread count.	Agent 13	2d
AGENT-04 — Financial Modeling Enhancements				
P2-11	**Live risk-free rate injection**	Pull FRED FEDFUNDS + GS10 from market_data_cache. Show suggested discount_rate on engagement settings. One-click apply.	Agent 4	1d
P2-12	**EBITDA bridge narrative**	POST /api/engagements/:id/ebitda-narrative. Claude reads bridge steps, writes 150-word management narrative. Cache 24h.	Agent 4	1d
P2-13	**Live HTS tariff lookup**	USITC DataWeb API for each spend record with country_of_origin. Compare live rate to static. Flag	Agent 4	2d

ID	Task	Description / Acceptance Criteria	Agent	Effort
		where diff > 2pp.		

PHASE 3 — Autonomous Operation (Weeks 11–20)

Goal: MD portfolio dashboard · Zero-touch pipeline · Client portal · Category advisor · Cross-engagement benchmarks

ID	Task	Description / Acceptance Criteria	Agent	Effort
AGENT-09 — MD Portfolio Command Center				
P3-01	**Portfolio snapshot nightly job**	Aggregate across all active engagements into portfolio_snapshots: total pipeline $, realized $, avg savings rate, at-risk count, team utilization. Anonymized peer benchmarks by industry + company_size.	Agent 9	2d
P3-02	**MD portfolio dashboard**	New home page /portfolio. KPI cards, engagement list, pipeline health, Claude weekly Portfolio Pulse digest.	Agent 9	3d
P3-03	**Peer benchmarking**	Cross-engagement savings rate by industry/size/lever (anonymized, min 5 engagements). 'Your 8.2% vs peer median 6.5%.' On initiative scoring and assumptions pages.	Agent 9	2d
Full Agentic Pipeline				
P3-04	**Zero-touch ingestion pipeline**	Upload → Agent 0 auto-enqueues: normalize → categorize → cleanse → analyze → score → Kraljic → exec brief. All via Agent 8 queue. <5 min for 50k records. SearXNG used for ambiguous supplier lookups during categorization.	Agents 0,1,2,8	3d
P3-05	**Client portal mode**	White-label toggle per engagement. PE sponsor login (read-only). Configurable page visibility. Separate URL: /portal/:engagementId/dashboard.	Infra	3d
AGENTS 10–11 — Category Advisor + 100-Day Plan				
P3-06	**Category brief generator**	POST /api/engagements/:id/categories/:categoryId/brief. Claude reads Kraljic position, spend, vendor risk, market benchmarks. Generates 1-page Word category strategy brief. Exa.ai semantic search for competitor supplier intel ('Find suppliers of industrial gases similar to Air Products in Europe').	Agent 10	2d
P3-07	**Competitor supplier intel**	Exa.ai semantic call per category brief generation. Append market intelligence section with cited sources. Fallback to SearXNG if Exa.ai credit exhausted.	Agent 10	1d
P3-08	**100-Day plan enhancements**	Auto-sequencing by dependency + resource constraints. RACI matrix generation. Weekly status report auto-draft (Claude). Gantt export to pptx.	Agent 11	3d

SECTION 5: TECHNOLOGY STACK — UPDATED

All new dependencies from Revision 2 additions highlighted

Category	Technology	Version	Purpose	Auth	Status
AI — Primary	Anthropic Claude API	claude-sonnet-4-20250514	ALL synthesis, narrative, risk classification, extraction, co-pilot, deliverable authoring	You have the key	No change
Search — Primary	SearXNG	Latest Docker image	Self-hosted metasearch. 70+ engines, unlimited, $0. Primary web search for all agents.	No key (self-hosted)	**NEW — replaces Perplexity**
Search — Fallback	DDGS Python lib	`pip install ddgs`	DuckDuckGo + multi-engine fallback when SearXNG rate-limited. Also ddgs.news() for news.	None	**NEW — replaces Perplexity**
Search — Reserve	Serper.dev	REST API	Google-quality results reserved for high-value queries. 2,500 free, then $1/1k.	Register: serper.dev	**NEW — replaces Perplexity**
Search — Semantic	Exa.ai	REST API	Neural semantic search for category intel and competitor research. $10 no-expiry credit.	Register: exa.ai	**NEW — replaces Perplexity**
News — Primary	NewsData.io	REST API	2,000 article URLs/day free, resets daily. 77 languages, 151 countries.	Register: newsdata.io	**NEW — replaces NewsAPI.ai**
News — Full Text	Trafilatura	`pip install trafilatura`	Extracts full article text from URLs. Apache 2.0. #1 ranked open-source extractor.	None	**NEW — solves full-text gap**
News — Trade Journals	feedparser	`pip install feedparser`	Parses RSS feeds from Supply Chain Dive, Spend Matters, FreightWaves, Reuters.	None	**NEW**
News — Fallback	DDGS news	`pip install ddgs`	ddgs.news() for real-time supplier news when NewsData.io daily credit exhausted.	None	**NEW**
Commodities	yfinance	`pip install yfinance`	20+ futures tickers (GC=F, HG=F, CL=F, etc.) in one call. No key, no limit.	None	**NEW — replaces CommodityPriceAPI**
Economics	FRED API	REST API	840k+ series. CPI, PPI by commodity, Fed Funds, Treasury, ECI.	Register: fred.stlouisfed.org	No change
Labor	BLS API v2	REST API	OES wages by occupation + geography, CPI, QCEW employment cost.	Register: bls.gov	No change
Energy	EIA API	REST API	WTI, nat gas, electricity. Authoritative US energy price source.	Register: eia.gov	No change
Commodities — Global	World Bank Pink Sheet	Monthly xlsx download	70+ commodities, 60yr history. Auto-download + parse with openpyxl.	None	No change
FX Rates	ECB FX API	REST API	40+ currency pairs to EUR. TTL 4h.	None	No change
Company Data	SEC EDGAR	REST API	10-K/10-Q financials for public suppliers. Altman Z-score.	None	No change
Company Data	OpenCorporates	REST API	200M+ company records. Ownership + officers. 50 calls/day free.	Register	No change
Sanctions	OFAC SDN	Bulk XML download	US Treasury sanctions list. Weekly refresh.	None	No change
Debarment	SAM.gov	REST API	US federal entity exclusions and registration.	Register	No change
Tariffs	USITC DataWeb	REST API	Live HTS tariff rates by product code and country.	Account required	No change
Supplier News (archive)	archive.ph	URL pattern	Paywall fallback for Trafilatura. https://archive.ph/{url} — no API needed.	None	**NEW**
Backend	Node.js + Express	v20 LTS	API server, route handlers, middleware.	N/A	No change

Category	Technology	Version	Purpose	Auth	Status
Database	SQLite + Drizzle ORM	better-sqlite3 8.x	Embedded DB + type-safe queries.	N/A	No change
Job Queue	BullMQ + Redis	bullmq 5.x + ioredis	Async agent task queue.	N/A	No change
Scheduling	node-cron	3.x	Nightly market refresh, portfolio snapshot, news scans.	N/A	No change
Python Sidecar	FastAPI	0.110+	Port 5001. Generates pptx/docx/xlsx/pdf. Hosts Trafilatura + feedparser endpoints.	N/A	No change
Doc Gen	python-pptx + python-docx + openpyxl + reportlab	Latest	Steerco decks, ODD memos, Excel models, PDF snapshots.	N/A	No change
Contract Parse	pdf-parse + mammoth	`npm packages`	PDF and docx contract text extraction.	N/A	No change
Web Scraping	playwright	1.40+	Drewry WCI weekly freight index scraper.	N/A	No change
Streaming	Server-Sent Events	Native Node.js	Job progress + co-pilot streaming.	N/A	No change

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28 tables · 14 agents · 4 search sources · 5 news sources · 4 commodity sources · 3 phases · 39 tasks · $0/month on data